ELECTRIC CHARGES AND FIELDS

Class 12 Physics

1. Introduction to Electric Charges

Electrostatics is the study of forces, fields, and potentials arising from static charges. The concept of electric charge originated from observations of phenomena like amber attracting light objects after rubbing. Experiments show that there are two kinds of electric charges, conventionally termed positive and negative. Objects acquire charge through friction, conduction, or induction.

Like charges repel each other, while unlike charges attract each other. When charged bodies are brought into contact, charges can neutralize, leading to electrical neutrality. The SI unit of electric charge is the Coulomb (C).

- Charge is a fundamental property of matter.
- Positive charge: glass rod rubbed with silk.
- Negative charge: plastic rod rubbed with fur.

2. Conductors and Insulators

Materials are classified based on their ability to allow electric charges to move through them.

Conductors: These materials allow electric charge to flow freely through them. They possess a large number of 'free electrons' that are loosely bound to their atoms and can move throughout the material. Examples include metals (e.g., copper, aluminum), the human body, and the Earth.

Insulators: These materials do not allow electric charge to flow freely. Their electrons are tightly bound to their atoms and are not free to move. When an insulator is charged, the charge remains localized at the point of application. Examples include glass, plastic, wood, and rubber.

Semiconductors: These materials have electrical conductivity between that of conductors and insulators. Their conductivity can be controlled by factors like temperature or the addition of impurities (doping). Examples are silicon and germanium.

- Conductors have free charge carriers.
- Insulators have tightly bound electrons.
- Semiconductors have controllable conductivity.

3. Charging by Induction

Charging by induction is a process by which a conductor can be charged without actually touching a charged object. When a charged body is brought near an uncharged conductor, the free electrons in the conductor redistribute themselves due to attraction or repulsion from the external charge. For instance, if a positively charged rod is brought near a neutral sphere, electrons in the sphere are attracted to the side nearer the rod, leaving the far side positively charged.

If the sphere is then grounded (connected to the Earth), the excess charge on the far side (positive, in this example) is neutralized by electrons flowing from the Earth. After the ground connection is removed and the charged rod is taken away, the sphere is left with a net negative charge, opposite to that of the inducing charge. This process ensures no direct transfer of charge from the inducing body to the induced body.

- Charges redistribute in a neutral conductor when a charged object is nearby.
- No direct contact is required for charging.
- The induced charge is always opposite in sign to the inducing charge.

4. Basic Properties of Electric Charge

Electric charge exhibits several fundamental properties:

4.1. Additivity of Charges: The total electric charge of a system is the algebraic sum of all individual charges present within the system. If a system contains charges $q_1, q_2, q_3, \ldots, q_n$, the total charge Q is given by $Q = q_1 + q_2 + q_3 + \cdots + q_n$.

4.2. Quantization of Charge: Electric charge is always an integral multiple of a fundamental unit of charge, denoted by 'e'. This smallest unit of charge is the magnitude of charge on an electron or a proton, approximately 1.602×10^{-19} C. Thus, any charge Q can be expressed as $Q = \pm ne$, where n is an integer $(1, 2, 3, \ldots)$. The existence of fractional charges (quarks) within elementary particles is noted, but they do not exist freely.

4.3. Conservation of Charge: For an isolated system, the total electric charge remains constant. Charge can neither be created nor destroyed; it can only be transferred from one body to another. For example, when a glass rod is rubbed with silk, electrons are transferred from the rod to the silk. The positive charge gained by the rod equals the negative charge gained by the silk, keeping the net charge of the rod-silk system zero.

- Charge is additive (algebraic sum).
- Charge is quantized ($Q = \pm ne$).
- Charge is conserved in an isolated system.

5. Coulomb's Law

Coulomb's Law describes the electrostatic force between two stationary point charges. A 'point charge' is an idealized charge concentrated at a single point in space. The law states that the force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them. The force acts along the line joining the two charges.

For two point charges q_1 and q_2 separated by a distance r in vacuum, the magnitude of the force F is given by:

$$F = k\frac{|q_1 q_2|}{r^2}$$

Where:

* k is the electrostatic constant or Coulomb's constant, $k = \frac{1}{4\pi\epsilon_0}$.

* ϵ_0 (epsilon naught) is the permittivity of free space, approximately $8.854 \times 10^{-12}\ \mathrm{C^2\ N^{-1}\ m^{-2}}$.

* The value of $k \approx 9 \times 10^9\ \mathrm{N\ m^2\ C^{-2}}$.

Vector Form of Coulomb's Law:

If $\vec{r}_{12}$ is the position vector from q_1 to q_2, then the force on q_2 due to q_1 is:

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}\hat{r}_{12}$$

Where $\hat{r}_{12}$ is the unit vector pointing from q_1 to q_2.

* If $q_1 q_2 > 0$ (like charges), the force is repulsive.

* If $q_1 q_2 < 0$ (unlike charges), the force is attractive.

- Force is proportional to product of charges.
- Force is inversely proportional to square of distance.
- Acts along the line connecting the charges.

6. Forces between Multiple Charges (Superposition Principle)

When there are multiple point charges, the net force on any one charge is the vector sum of the individual forces exerted on that charge by all the other charges. This is known as the 'Principle of Superposition'. The presence of other charges does not affect the force between any two given charges.

If a system consists of n charges $q_1, q_2, \ldots, q_n$, and we want to find the total force on charge q_1, then:

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \cdots + \vec{F}_{1n}$$

Where $\vec{F}_{1i}$ is the force on charge q_1 due to charge q_i, calculated using Coulomb's Law, considering only q_1 and q_i at a time. Each force is a vector, so their sum must be a vector sum, accounting for both magnitude and direction.

- Net force is the vector sum of individual forces.
- Forces between pairs of charges are independent.
- Applies to any number of interacting charges.

7. Electric Field

An electric field is a region of space around a charged object where another charged object would experience an electrostatic force. It is a vector quantity.

Definition: The electric field $\vec{E}$ at a point is defined as the electrostatic force $\vec{F}$ experienced by a unit positive test charge q_0 placed at that point, such that q_0 is small enough not to disturb the source charge distribution.

$$\vec{E} = \frac{\vec{F}}{q_0}$$

The SI unit of electric field is Newton per Coulomb (N/C).

Electric Field due to a Point Charge:

For a single point charge Q at the origin, the electric field $\vec{E}$ at a distance r from the charge is given by:

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\hat{r}$$

Where $\hat{r}$ is the unit vector pointing from the source charge Q to the point where the field is calculated. If Q is positive, $\vec{E}$ points radially outwards; if Q is negative, $\vec{E}$ points radially inwards.

Principle of Superposition for Electric Fields:

The electric field at any point due to a system of charges is the vector sum of the electric fields produced by each individual charge at that point.

$$\vec{E}_{total} = \vec{E}_1 + \vec{E}_2 + \cdots + \vec{E}_n$$

- Region where a test charge experiences force.
- Vector quantity (magnitude and direction).
- Follows superposition principle for multiple charges.

8. Electric Field Lines

Electric field lines are a visual tool to represent electric fields. They are imaginary lines or curves drawn such that the tangent to the curve at any point gives the direction of the electric field at that point. The density of the lines (number of lines per unit area perpendicular to the lines) is proportional to the magnitude of the electric field.

Properties of Electric Field Lines:

* **Origin and Termination:** Field lines originate from positive charges and terminate on negative charges. If there's a single charge, they extend to infinity.

* **No Intersection:** Two field lines can never intersect. If they did, it would imply two directions for the electric field at the point of intersection, which is impossible.

* **Continuous Curves:** They are continuous curves in a charge-free region, without sudden breaks.

* **No Closed Loops:** Electrostatic field lines do not form closed loops because the electrostatic field is conservative (work done in moving a charge along a closed path is zero).

* **Perpendicular to Conductor Surface:** They are always perpendicular to the surface of a conductor, both when starting from or ending on it.

* **Crowding and Spacing:** Where lines are closer, the field is stronger; where they are farther apart, the field is weaker.

* **Do not Pass Through a Conductor:** Inside a conductor in electrostatic equilibrium, the electric field is zero, so field lines do not pass through it.

- Visualize electric field direction and strength.
- Originate from positive, end on negative charges.
- Never cross each other; do not form closed loops.

9. Electric Flux

Electric flux is a measure of the 'flow' of electric field through a given surface. It quantifies the number of electric field lines passing through an area. It is a scalar quantity.

Definition: The electric flux $\Delta\Phi_E$ through an infinitesimal area element $d\vec{S}$ in an electric field $\vec{E}$ is given by the dot product:

$$\Delta\Phi_E = \vec{E} \cdot d\vec{S} = EdS\cos\theta$$

Where θ is the angle between the electric field vector $\vec{E}$ and the area vector $d\vec{S}$ (which is a vector normal to the surface, pointing outwards).

For a finite surface, the total electric flux Φ_E is found by integrating over the entire surface:

$$\Phi_E = \int_S \vec{E} \cdot d\vec{S}$$

For a closed surface, the integral is written as $\oint_S \vec{E} \cdot d\vec{S}$.

The SI unit of electric flux is Newton-meter squared per Coulomb (N m^2 C^{-1}) or Volt-meter (V m). Positive flux indicates field lines leaving the surface, while negative flux indicates lines entering.

- Measure of electric field lines passing through a surface.
- Scalar quantity, calculated as a dot product.
- Unit: N m^2 C^{-1} or V m.

10. Electric Dipole

An electric dipole is a pair of equal and opposite point charges, $+q$ and $-q$, separated by a small fixed distance, typically denoted as $2a$.

Electric Dipole Moment ($\vec{p}$): This is a vector quantity that characterizes the strength and orientation of an electric dipole. Its magnitude is the product of the magnitude of either charge and the distance between them.

$$|\vec{p}| = q \times 2a$$

The direction of the electric dipole moment vector is conventionally from the negative charge to the positive charge.

Electric Field due to an Electric Dipole:

10.1. On the Axial Line (at point P): For a point P on the axis of the dipole at a distance r from its center ($r \gg a$), the electric field $\vec{E}_{axial}$ is in the same direction as $\vec{p}$.

$$\vec{E}_{axial} = \frac{1}{4\pi\epsilon_0}\frac{2\vec{p}}{r^3}$$

10.2. On the Equatorial Line (at point P): For a point P on the equatorial plane (perpendicular bisector) of the dipole at a distance r from its center ($r \gg a$), the electric field $\vec{E}_{equatorial}$ is in the direction opposite to $\vec{p}$.

$$\vec{E}_{equatorial} = -\frac{1}{4\pi\epsilon_0}\frac{\vec{p}}{r^3}$$

- Pair of equal and opposite charges separated by a small distance.
- Dipole moment $\vec{p}$ from $-q$ to $+q$.
- Field on axis is parallel to $\vec{p}$, on equator is anti-parallel.

11. Dipole in a Uniform External Field

When an electric dipole is placed in a uniform external electric field $\vec{E}$, the two charges $+q$ and $-q$ experience forces of equal magnitude ($F = qE$) but in opposite directions. These two forces form a couple, which exerts a torque on the dipole.

Torque ($\vec{\tau}$): The torque experienced by the electric dipole tends to align it with the external electric field. The magnitude of this torque is given by:

$$\vec{\tau} = \vec{p} \times \vec{E}$$

The magnitude of the torque is $\tau = pE\sin\theta$, where θ is the angle between the dipole moment $\vec{p}$ and the electric field $\vec{E}$.

* When $\theta = 0°$ (dipole aligned with the field), $\tau = 0$, representing stable equilibrium.

* When $\theta = 180°$ (dipole anti-aligned with the field), $\tau = 0$, representing unstable equilibrium.

* When $\theta = 90°$, $\tau = pE$, which is the maximum torque.

No net force acts on the dipole in a uniform electric field, as the forces on $+q$ and $-q$ are equal and opposite, thus cancelling out. However, if the field is non-uniform, there will be a net force as well as a torque.

- Uniform field applies torque, no net force.
- Torque tends to align dipole with the field.
- Maximum torque when $\vec{p}$ is perpendicular to $\vec{E}$.

12. Continuous Charge Distribution

In many practical situations, charges are distributed continuously over a line, surface, or volume rather than being discrete point charges. To calculate the electric field due to such distributions, we consider infinitesimal charge elements and integrate their contributions. This is a crucial step for real-world scenarios.

12.1. Linear Charge Density (λ): If charge is distributed uniformly along a line, the linear charge density is the charge per unit length.

$$\lambda = \frac{dQ}{dL} \quad (\text{Unit: C/m})$$

Where dQ is the charge on an infinitesimal length dL.

12.2. Surface Charge Density (σ): If charge is distributed uniformly over a surface, the surface charge density is the charge per unit area.

$$\sigma = \frac{dQ}{dA} \quad (\text{Unit: C/m}^2)$$

Where dQ is the charge on an infinitesimal area dA.

12.3. Volume Charge Density (ρ): If charge is distributed uniformly throughout a volume, the volume charge density is the charge per unit volume.

$$\rho = \frac{dQ}{dV} \quad (\text{Unit: C/m}^3)$$

Where dQ is the charge in an infinitesimal volume dV.

The electric field due to a continuous charge distribution is found by integrating the electric field contributions from all infinitesimal charge elements: $\vec{E} = \int d\vec{E}$, where $d\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{dQ}{r^2}\hat{r}$.

- Charge spread over a line, surface, or volume.
- Use linear (λ), surface (σ), or volume (ρ) charge densities.
- Electric field is found by integration over infinitesimal charge elements.

13. Gauss's Law

Gauss's Law is a fundamental law in electromagnetism that relates the electric flux through any closed surface (called a Gaussian surface) to the net electric charge enclosed within that surface. It is one of Maxwell's equations and is particularly powerful for calculating electric fields for charge distributions with high symmetry.

Statement: The total electric flux Φ_E through any closed surface in vacuum is equal to $\frac{1}{\epsilon_0}$ times the total (net) electric charge $Q_{enclosed}$ enclosed within that surface.

$$\Phi_E = \oint_S \vec{E} \cdot d\vec{S} = \frac{Q_{enclosed}}{\epsilon_0}$$

Where:

* $\oint_S$ denotes the integral over a closed surface S.

* $Q_{enclosed}$ is the net charge enclosed by the Gaussian surface.

* ϵ_0 is the permittivity of free space.

Key Points for Applying Gauss's Law:

1. **Symmetry:** Choose a Gaussian surface that matches the symmetry of the charge distribution (e.g., sphere for point charge, cylinder for line charge).

2. **Electric Field:** The electric field $\vec{E}$ should ideally be constant in magnitude and perpendicular to the Gaussian surface, or parallel to the surface (where $\vec{E} \cdot d\vec{S} = 0$).

3. **Enclosed Charge:** Carefully identify and sum all charges enclosed within the chosen Gaussian surface.

- Relates electric flux to enclosed charge.
- Useful for symmetric charge distributions.
- $\Phi_E = Q_{enclosed}/\epsilon_0$.

14. Applications of Gauss's Law

Gauss's Law significantly simplifies the calculation of electric fields for various symmetrical charge distributions:

14.1. Electric Field due to an Infinitely Long Straight Uniformly Charged Wire:

For an infinitely long straight wire with a uniform linear charge density λ, by choosing a cylindrical Gaussian surface coaxial with the wire, the electric field $\vec{E}$ at a radial distance r from the wire is radially outward (if $\lambda > 0$) and its magnitude is:

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

14.2. Electric Field due to a Uniformly Charged Infinite Plane Sheet:

For an infinite plane sheet with a uniform surface charge density σ, by choosing a cylindrical or rectangular Gaussian surface with its flat ends perpendicular to the sheet, the electric field $\vec{E}$ is uniform, perpendicular to the sheet, and its magnitude is:

$$E = \frac{\sigma}{2\epsilon_0}$$

The field points away from the sheet if $\sigma > 0$ and towards it if $\sigma < 0$.

14.3. Electric Field due to a Uniformly Charged Thin Spherical Shell:

Consider a thin spherical shell of radius R with a total charge Q uniformly distributed on its surface ($\sigma = Q/(4\pi R^2)$).

* **Outside the shell ($r > R$):** The electric field $\vec{E}$ at a distance r from the center is the same as if all the charge Q were concentrated at the center of the shell (like a point charge).

$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$$

* **Inside the shell ($r < R$):** Since no charge is enclosed within a Gaussian surface inside the shell ($Q_{enclosed} = 0$), the electric field inside a uniformly charged spherical shell is zero.

$$E = 0$$

- Simplifies field calculation for symmetric charge distributions.
- Wire: $E \propto 1/r$.
- Plane sheet: E is uniform, independent of distance.
- Spherical shell: $E = 0$ inside, like point charge outside.